

Java Collections Code Examples

أمثلة أكواد هياكل البيانات في الجافا

1. ArrayList

```
import java.util.ArrayList;

public class ArrayListExample {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();

        list.add("B");
        list.add(0, "A");
        list.add("C");

        System.out.println(list.get(1));

        list.set(1, "Updated_B");

        list.remove(0);

        System.out.println(list);
    }
}
```

2. LinkedList

```
import java.util.LinkedList;

public class LinkedListExample {
    public static void main(String[] args) {
        LinkedList<String> list = new LinkedList<>();

        list.add("B");
        list.addFirst("A");
        list.addLast("C");

        System.out.println(list.get(1));

        list.set(1, "Updated_B");

        list.removeFirst();

        System.out.println(list);
    }
}
```

3. Vector

```
import java.util.Vector;

public class VectorExample {
    public static void main(String[] args) {
        Vector<String> vector = new Vector<>();

        vector.add("B");
        vector.add(0, "A");
        vector.add("C");

        System.out.println(vector.get(1));

        vector.set(1, "Updated_B");

        vector.remove(0);

        System.out.println(vector);
    }
}
```

4. Stack

```
import java.util.Stack;

public class StackExample {
    public static void main(String[] args) {
        Stack<String> stack = new Stack<>();

        stack.push("A");
        stack.add(1, "B");
        stack.push("C");

        System.out.println(stack.peek());

        stack.set(1, "Updated_B");

        stack.pop();

        System.out.println(stack);
    }
}
```

5. ArrayDeque

```
import java.util.ArrayDeque;

public class ArrayDequeExample {
    public static void main(String[] args) {
        ArrayDeque<String> deque = new ArrayDeque<>();
    }
}
```

```

        deque.add("B");
        deque.addFirst("A");
        deque.addLast("C");

        System.out.println(deque.peekFirst());

        deque.remove("B");
        deque.add("Updated_B");

        deque.removeFirst();

        System.out.println(deque);
    }
}

```

6. HashSet

```

import java.util.HashSet;

public class HashSetExample {
    public static void main(String[] args) {
        HashSet<String> set = new HashSet<>();

        set.add("Apple");
        set.add("Banana");
        set.add("Apple");
        set.add("Orange");

        System.out.println(set);

        set.remove("Banana");
        set.add("Mango");

        set.remove("Apple");

        System.out.println(set);
    }
}

```

7. PriorityQueue

```

import java.util.PriorityQueue;

public class PriorityQueueExample {
    public static void main(String[] args) {
        PriorityQueue<String> pq = new PriorityQueue<>();

        pq.add("Banana");
        pq.add("Apple");
    }
}

```

```
    pq.add("Cherry");

    System.out.println(pq.peek());

    pq.remove("Banana");
    pq.add("Durian");

    pq.poll();

    System.out.println(pq);
}
```